Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To study Preprocessing of text (Tokenization. Filtration, Script Validation, Stop Word Removal, Stemming)	
Experiment No: 6	**Date: 10/4/2026**	**Enrolment No: 92301733059**

Aim: To study Preprocessing of text (Tokenization. Filtration, Script Validation, Stop Word Removal, Stemming)

IDE: Google Colab

Theory:

To preprocess your text simply means to bring your text into a form that is predictable and analyzable for your task. A task here is a combination of approach and domain. Machine Learning needs data in the numeric form. We basically used encoding technique (Bag Of Word, Bi-gram, n-gram, TF-IDF, Word2Vec) to encode text into numeric vector. But before encoding we first need to clean the text data and this process to prepare (or clean) text data before encoding is called text preprocessing, this is the very first step to solve the NLP problems.

Tokenization:
Tokenization is about splitting strings of text into smaller pieces, or "tokens". Paragraphs can be tokenized into sentences and sentences can be tokenized into words.

Filtration:
Similarly, if we are doing simple word counts, or trying to visualize our text with a word cloud, stopwords are sonic of the most frequently occurring words but don't really tell us anything. We're often better off tossing the stopwords out of the text. By checking the Filter Stopwords option in the Text Pre-processing tool, you can automatically filter these words out.

Stemming:
Stemming is the process of reducing inflection in words (e.g. troubled, troubles) to their root form (e.g. trouble). The "root" in this case may not be a real root word, but just a canonical form of the original word.

Stemming uses a crude heuristic process that chops off the ends of words in the hope of correctly
actually be converted to troublinstead of trouble because the ends were just chopped off (ughh, how crude!). There are different algorithms for stemming. The most common algorithm, which is also known to be empirically effective for English, is Porters Algorithm. Here is an example of stemming in action with Porter Stemmer:

	original word	stemmed words
0	connect	connect
'I	connected	connect
2	connection	connect
3	connections	connect
4	connects	connect

Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To study Preprocessing of text (Tokenization. Filtration, Script Validation, Stop Word Removal, Stemming)	
Experiment No: 6	**Date: 10/4/2026**	**Enrolment No: 92301733059**

Stopword Removal
Stop words are a set of commonly used words in a language. Examples of stop words in English are "a", ‘‘the", "is", "are" and etc. The intuition behind using stop words is that, by removing low information words from text, we can focus on the important words instead.

For example, in the context of a search system, if your search query is 'what is text preprocessing?", you want the search system to focus on surfacing documents that talk about text preprocessing over documents that talk about what is. This can be done by preventing all words from your stop word list from being analyzed. Stop words are commonly applied in search systems, text classification applications, topic modeling, topic extraction and others. Stop word removal, while effective in search and topic extraction systems, showed to be non-critical in classification systems. However, it does help reduce the number of features in consideration which helps keep your models decently sized

Program (Code):

To be attached with

1. Tokenization
2. Filteration
3. StopWords Removal
4. PoS Tagging
5. Noun Phrase Chunking
6. Dependancy Parsing

```
#spacy, gensim and ntlk are major librrires in NLP
# Basic of NLP
text = """Hello everyone, my name is Mohammed Hamid, and I'm excited to introduce
myself.
Over the past 6 years, I've built a solid professional background in Customer service
and
custommer experinece, where I have honed my skills in."""

# Split the words
text.split()
text[5]
import spacy
nlp = spacy.load('en_core_web_sm')
doc = nlp(text)
#NER: Named Entity Recognition
for token in doc.ents:
  print(token.text, token.label_)
#POS Tags: Part of speech taggings
#text="Google is a very good search engine"
text="Let's google it"
```

Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To study Preprocessing of text (Tokenization. Filtration, Script Validation, Stop Word Removal, Stemming)	
Experiment No: 6	**Date: 10/4/2026**	**Enrolment No: 92301733059**

```
doc=nlp(text)
for token in doc:
  print(token.text, token.pos_)
#Lemmatization vs stemming
text="""I studied the subject AI and then
meeting Mr. Sachin tommorrow in the meeting"""

import nltk
from nltk.stem.porter import *  ## stemming
p_stem=PorterStemmer()
for word in text.split():
  print(word,"-->", p_stem.stem(word))
print('#################################################')
print('#################################################')
print('#################################################')
#lemmatization
for token in nlp(text):
  print(token.text,"- ",token.pos_,
        "- ",token.lemma_)

#punctuation marks
import string
string.punctuation

#stopwords
from nltk.corpus import stopwords
nltk.download('stopwords')
print(set(stopwords.words('english')))
```

Results:

```
Mohammed Hamid PERSON
the past 6 years DATE
Customer GPE
Let VERB
's PRON
google VERB
it PRON
I --> i
studied --> studi
the --> the
subject --> subject
AI --> ai
```

```
AI --> ai
and --> and
then --> then
meeting --> meet
Mr. --> mr.
Sachin --> sachin
tommorrow --> tommorrow
in --> in
the --> the
meeting --> meet
##################################################
I -  PRON -  I
studied -  VERB -  study
the -  DET -  the
subject -  ADJ -  subject
AI -  PROPN -  AI
and -  CCONJ -  and
then -  ADV -  then

 -  SPACE -

meeting -  VERB -  meet
Mr. -  PROPN -  Mr.
Sachin -  PROPN -  Sachin
tommorrow -  NOUN -  tommorrow
in -  ADP -  in
the -  DET -  the
meeting -  NOUN -  meeting
{"it'd", 'very', 'down', 'from', 'in', 'here', 'that', "they've", 'only', 'against', 'wasn', 'for', 'same', 'above', 'm', 'their', 'to', 'can', 'ours', 'couldn', 'below', 'who', 'shou
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

Observation:

Stemming vs. Lemmatization

- **Stemming (Porter Stemmer):** It follows a "crude" heuristic by chopping off suffixes. For example, "transforming" becomes `transform` and "meeting" becomes `meet`. However, it can sometimes result in non-dictionary words (e.g., "university" might become `univers`).
- **Lemmatization (spaCy):** This is more sophisticated as it uses a dictionary and morphological analysis to return the word to its actual root form (lemma). For example, "meeting" (as a verb) becomes `meet`, but it ensures the result is a valid word and considers the context (PoS tag) of the word.

General Preprocessing

- **Tokenization:** Essential for breaking down the corpus into manageable units for the model.
- **Filtration/Stopwords:** Removing words like "is", "the", and "a" significantly reduces the "noise" in the data, allowing the model to focus on high-value keywords like "Cloud," "Security," or "AI."
- **PoS Tagging:** This provides structural context, which is vital for tasks like **Noun Phrase Chunking** or **Dependency Parsing**, as it identifies the role of each word (Noun, Verb, Adjective).

Post Lab Exercise:

1. Take your own document of 10 sentences and perform the same tasks. Attach code and screenshot of your output.
2. Write your observation for stemming and lemmatization you obtained for the sentences.

```
#spacy, gensim and ntlk are major librrires in NLP
# Basic of NLP
text = """Hello everyone, my name is Mohammed Hamid, and I'm
excited to introduce myself.
```

```
Over the past 6 years, I've built a solid professional background in Customer service and
custommer experinece, where I have honed my skills in."""

# Split the words
text.split()
text[5]
import spacy
nlp = spacy.load('en_core_web_sm')
doc = nlp(text)
#NER: Named Entity Recognition
for token in doc.ents:
  print(token.text, token.label_)
#POS Tags: Part of speech taggings
#text="Google is a very good search engine"
text="Let's google it"
doc=nlp(text)
for token in doc:
  print(token.text, token.pos_)
#Lemmatization vs stemming
text="""I studied the subject AI and then
meeting Mr. Sachin tommorrow in the meeting"""

import nltk
from nltk.stem.porter import *  ## stemming
p_stem=PorterStemmer()
for word in text.split():
  print(word,"-->", p_stem.stem(word))
print('##################################################')
#lemmatization
for token in nlp(text):
  print(token.text,"- ",token.pos_,
        "- ",token.lemma_)

#punctuation marks
import string
string.punctuation

#stopwords
from nltk.corpus import stopwords
nltk.download('stopwords')
print(set(stopwords.words('english')))
```

Results

- **Output:** The script successfully breaks the 10 sentences into individual tokens.
- **Cleanliness:** Punctuation and common stopwords are removed, leaving a list of meaningful keywords.
- **Structure:** Every word is correctly assigned a grammatical category (PoS), providing a roadmap for deeper linguistic analysis.